A semi-Analytic Framework for Long–Short Position Sizing in DeFi under Liquidation and Jump Risk

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DeFi long–short sizing under liquidation and jump risk

DeFi Lending Platforms

Aave / Compound: pooled lending

- **Mechanism** lenders supply pool assets; borrowers draw pooled liquidity
- **Accounting** no bilateral lender; protocol-level collateral and debt balances
- **Interest accrual** variable supply/borrow APRs; utilization-driven index growth
- **No fixed tenor** open-ended loans; no maturity date; no term structure contract

Overcollateralization

- Pseudonymous parties; no credit underwriting; no legal margin call
- Solvency and obligation enforcement entirely collateral-based
- No-arbitrage discipline through collateral constraint
- **Liquidator role** repay unsolvent debt; seize collateral bonus; restore pool solvency

Health Factor and Boundary

Key mechanism variables

State variables:

$$V_{1,t} := n_1 S_{1,t} \quad \text{collateral value,} \qquad V_{2,t} := n_2 S_{2,t} \quad \text{borrow/debt value}$$

Loan-to-value* and liquidation threshold:

$$\mathrm{LTV}_t := \frac{V_{2,t}}{V_{1,t}}, \qquad b = \text{threshold LTV}$$

Aave origination LTV $<$ threshold LTV b; initial buffer by design

Health factor:

$$H_t := \frac{b\, V_{1,t}}{V_{2,t}} = \frac{b}{\mathrm{LTV}_t}$$

Liquidation triggering condition:

$$H_t < 1 \quad \Longleftrightarrow \quad V_{2,t} > b\, V_{1,t}$$

* Remark: for now, ignore origination-LTV / threshold-LTV gap

Constructing the Long–Short Position

Step	Protocol action	Economic exposure
0	Initial capital injection: \$1-worth BTC	External capital only
1	Supply BTC collateral to Aave	Long BTC collateral
2	Borrow \$0.8-worth ETH	Short ETH liability
3	Sell ETH into \$0.8-worth BTC	Self-financing conversion
4	Re-supply BTC as collateral	Total BTC collateral: \$1.8
Result	Collateralized debt position	Leveraged long BTC, short ETH

Resulting normalized position:

- External capital: \$1; no additional capital after step 0
- BTC collateral: \$1.8; ETH debt: \$0.8
- $b = 0.8$; $H_0 = 0.8 \times 1.8/0.8 = 1.8$
- Leveraged BTC exposure; short ETH exposure

Looping the Position

Repeat steps 1–4; zero transaction cost and slippage

Round-m additional collateral:

$$1, \quad b, \quad b^2, \quad \ldots, \quad b^m$$

Total BTC collateral after m loops:

$$V_1^{(m)} = 1 + b + \cdots + b^m = \frac{1 - b^{m+1}}{1 - b}$$

ETH debt:

$$V_2^{(m)} = b + b^2 + \cdots + b^{m+1} = b\, V_1^{(m)}$$

Infinite-loop limit:

$$V_1^{(\infty)} = \frac{1}{1 - b}, \qquad V_2^{(\infty)} = \frac{b}{1 - b}$$

Example $b = 0.8$:

$$V_1^{(\infty)} = 5, \qquad V_2^{(\infty)} = 4, \qquad H_0^{(\infty)} = \frac{b V_1^{(\infty)}}{V_2^{(\infty)}} = 1$$

Extreme leverage: any adverse move $\rightarrow$ liquidation; favorable move $\rightarrow$ amplified gain

Unwinding and Taking Profit

Favorable scenario: BTC $+10\%$; ETH -16.7%

$$H_t = \frac{0.80\, V_{1,t}}{V_{2,t}}, \qquad \text{equity} = V_{1,t} - V_{2,t}$$

Scenario	BTC collat.	ETH debt	H	Equity	Action
Entry	\$1.80	\$0.80	1.80	\$1.00	Hold
Favorable move	\$1.98	\$0.667	2.38	\$1.313	Voluntary unwind

Outcome:

$$\text{gain} = 1.313 - 1.000 = 0.313$$

Voluntary unwind; solvent position; profit realization

Aave Liquidation and Immediate Risk

Adverse scenario: BTC -25%; ETH $+40\%$

$$H_t = \frac{bV_{1,t}}{V_{2,t}} = \frac{0.80\, V_{1,t}}{V_{2,t}}, \qquad H_t \le 1 \iff \frac{V_{2,t}}{V_{1,t}} \ge 0.80$$

Scenario	BTC collat.	ETH debt	H	Equity	Outcome
Entry	\$1.80	\$0.80	1.80	\$1.00	Safe
Adverse move	\$1.35	\$1.12	0.96	\$0.23	Liquidatable

Outcome:

$$H_t < 1 \quad \Rightarrow \quad \text{keeper liquidation; collateral seized at bonus}$$

Recovery drivers: shock severity; market sentiment/liquidity; liquidator inventory willingness; gas fees

Modeling convention: detailed recovery omitted; $\Pi_T = 0$ whenever $\tau \le T$

Payoff Function from the Trade

Holdings: n_1 units of collateral asset S_1; debt: n_2 units of borrowed asset S_2

Surviving-path terminal wealth:

$$\Pi_T := (n_1 S_{1,T} - n_2 S_{2,T})\, \mathbf{1}_{\{\tau > T\}}$$

Running BTC/ETH example:

$$S_1 = \mathrm{BTC}, \qquad S_2 = \mathrm{ETH}, \qquad n_1 S_{1,0} = 1.80, \qquad n_2 S_{2,0} = 0.80$$

Therefore:

$$\Pi_T = \left(1.80\, \frac{S_{1,T}}{S_{1,0}} - 0.80\, \frac{S_{2,T}}{S_{2,0}}\right) \mathbf{1}_{\{\tau > T\}}$$

Survival indicator: $\tau \leq T \Rightarrow \Pi_T = 0$

Initial net book value normalized to one unit of account; moments stated directly in Π_T

Why Payoff Moments Are Hard

Target statistics:

$$\mathbb{E}[\Pi_T], \qquad \mathrm{Var}(\Pi_T), \qquad \mathbb{E}[\Pi_T^k], \quad k \geq 1$$

Killed payoff:

$$\Pi_T^k = (n_1 S_{1,T} - n_2 S_{2,T})^k \mathbf{1}_{\{\tau > T\}}$$

Terminal payoff state:

$$(S_{1,T}, S_{2,T})$$

Pathwise killing process:

$$\tau > T \quad \Longleftrightarrow \quad H_s \geq 1 \ \text{ for all } s \leq T, \qquad H_s = \frac{b\, n_1 S_{1,s}}{n_2 S_{2,s}}$$

Direct moment:

$$\mathbb{E}_{\mathbb{P}}[\Pi_T^k] = \mathbb{E}_{\mathbb{P}}\left[(n_1 S_{1,T} - n_2 S_{2,T})^k \mathbf{1}_{\{\tau > T\}}\right]$$

Difficulty: terminal bivariate payoff + pathwise barrier

Generic numerical routes:

- Monte Carlo; barrier monitoring / possible 2D Brownian-bridge refinement
- Numerical PDE/PIDE; two terminal state variables plus killing boundary

Literature Positioning

Intersection:

$$\text{DeFi liquidation} \quad + \quad \text{first-passage methods for jump diffusions}$$

- **DeFi liquidation** Qin et al. [4]; Heimbach–Huang [1]
- Incentive structure; leverage amplification; liquidation cascades
- Empirical focus; no ex-ante sizing rule

- **First passage under jumps** Kou–Wang [3]; Sepp [5]
- Double-exponential jumps; Laplace transforms; finite algebraic systems
- Classical object: single-asset barrier option

Gap:

$$\text{barrier-killed long–short ratio} \quad \neq \quad \text{single-asset barrier option}$$

Bridge: change of measure $\rightarrow$ one-dimensional killed expectation

Price Process

Jump motivation: crypto gaps; flash crashes; barrier crossed before rebalance — Kou model [2]

Price dynamics, $i = 1, 2$:

$$S_{i,t} = S_{i,0} e^{X_{i,t}}, \qquad dX_{i,t} = \mu_i^X \, dt + \sigma_i \, dB_{i,t} + d\left(\sum_{n=1}^{N_{i,t}} J_i^{(n)} \right)$$

Brownian correlation:

$$d\langle B_1, B_2 \rangle_t = \rho \, dt,$$

Poisson intensity: λ_i

Kou double-exponential jump density:

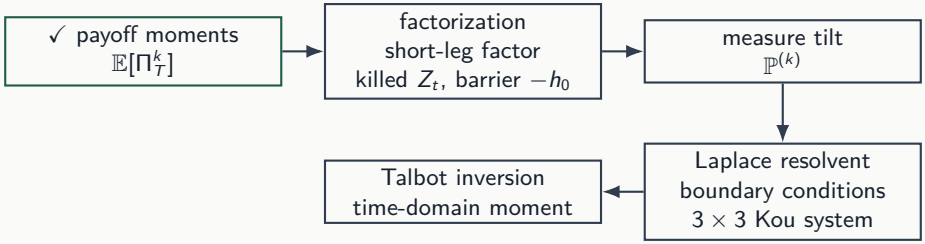
$$f_i(j) = \frac{p_i}{\eta_{i,+}} e^{-j/\eta_{i,+}} \mathbf{1}_{\{j>0\}} + \frac{1-p_i}{\eta_{i,-}} e^{j/\eta_{i,-}} \mathbf{1}_{\{j<0\}}$$

Drift normalization: $\mathbb{E}_{\P}[S_{i,t}/S_{i,0}] = e^{\mu_i t}$

$$\mu_i^X = \mu_i - \frac{1}{2}\sigma_i^2 - \lambda_i \chi_i, \qquad \chi_i = \mathbb{E}[e^{J_i} - 1] = \frac{p_i}{1-\eta_{i,+}} + \frac{1-p_i}{1+\eta_{i,-}} - 1$$

Constant supply/borrow rates absorbed into drift; dynamic rate modeling outside scope

Semi-Analytic Architecture



Covered: payoff function + need for moments

Next: factorization and measure change → one-dimensional killed expectation

$$k\,\eta_{2,+} < 1 \quad \text{moment-admissibility condition}$$

Factorization: From Two Assets to One Barrier

Motivation: separate terminal short-leg factor; isolate killed ratio problem

Bivariate payoff; scalar liquidation state:

$$Z_t = X_{1,t} - X_{2,t}$$

Initial log health factor:

$$h_0 := \log H_0$$

Factorization using $H_T = e^{h_0 + Z_T}$:

$$\Pi_T = \frac{n_2}{b} S_{2,0} e^{X_{2,T}} (e^{h_0 + Z_T} - b) \mathbf{1}_{\{\tau > T\}}$$

One-dimensional payoff kernel:

$$g_k(z; h_0) := (e^{h_0 + z} - b)^k,$$

Integer moments:

$$\Pi_T^k = \underbrace{\left(\frac{n_2}{b}\right)^k S_{2,0}^k e^{k X_{2,T}}}_{\mathcal{F}_T^{S_2} \text{ part}} \underbrace{g_k(Z_T; h_0) \mathbf{1}_{\{\tau > T\}}}_{\mathcal{F}_T^Z \text{ killed-ratio part}}$$

Measure change: absorb $\mathcal{F}_T^{S_2}$ factor; retain killed expectation under Z

Barrier-option analogue: scalar state Z, barrier $-h_0$

Measure Change

Start from factorized moment:

$$\mathbb{E}_{\mathbb{P}}[\Pi_T^k] = \left(\frac{n_2}{b}\right)^k S_{2,0}^k \, \mathbb{E}_{\mathbb{P}}\left[e^{kX_{2,T}} g_k(Z_T; h_0)\mathbf{1}_{\{\tau > T\}}\right]$$

Exponential tilt:

$$\left.\frac{d\mathbb{P}^{(k)}}{d\mathbb{P}}\right|_{\mathcal{F}_T} = \frac{e^{kX_{2,T}}}{\mathbb{E}[e^{kX_{2,T}}]}$$

Heuristic reminder, with $Y = g_k(Z_T; h_0)\mathbf{1}_{\{\tau > T\}}$:

$$\mathbb{E}_{\mathbb{P}^{(k)}}[Y] = \int Y \, d\mathbb{P}^{(k)} = \int Y \, \frac{d\mathbb{P}^{(k)}}{d\mathbb{P}} \, d\mathbb{P}$$

$$= \mathbb{E}_{\mathbb{P}}\left[Y \, \frac{e^{kX_{2,T}}}{\mathbb{E}_{\mathbb{P}}[e^{kX_{2,T}}]}\right] \qquad = \frac{\mathbb{E}_{\mathbb{P}}[e^{kX_{2,T}} Y]}{\mathbb{E}_{\mathbb{P}}[e^{kX_{2,T}}]} \quad \Longrightarrow \quad \mathbb{E}_{\mathbb{P}}[e^{kX_{2,T}} Y] = \mathbb{E}_{\mathbb{P}}[e^{kX_{2,T}}]\mathbb{E}_{\mathbb{P}^{(k)}}[Y]$$

Remark: Radon–Nikodym theorem / change-of-measure identity

Reduced moment:

$$\mathbb{E}[\Pi_T^k] = \left(\frac{n_2}{b}\right)^k S_{2,0}^k \mathbb{E}[e^{kX_{2,T}}] \, \mathbb{E}^{(k)}\left[g_k(Z_T; h_0)\mathbf{1}_{\{\tau > T\}}\right]$$

Bivariate payoff moment → killed expectation of the ratio process

Tilted Ratio Process

Remaining object:

$$\mathbb{E}^{(k)}\left[g_k(Z_T; h_0)\mathbf{1}_{\{\tau > T\}}\right]$$

Question:

$$Z_t = X_{1,t} - X_{2,t} \quad \text{under} \quad \mathbb{P}^{(k)}$$

Under $\mathbb{P}^{(k)}$: Z remains a Lévy process

Lévy–Khintchine exponent of Z under $\mathbb{P}^{(k)}$:

$$\psi_Z^{(k)}(s) = \mu_Z^{(k)} s + \frac{1}{2}\sigma_Z^2 s^2 + \lambda_1\big(M_1(s) - 1\big) + \lambda_2\big(M_2(k-s) - M_2(k)\big)$$

with

$$M_i(s) = \frac{p_i}{1 - s\eta_{i,+}} + \frac{1 - p_i}{1 + s\eta_{i,-}}, \qquad \sigma_Z^2 = \sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2$$

$\mathcal{L}^{(k)}$: infinitesimal generator of Z under $\mathbb{P}^{(k)}$, the generator eigenrelation:

$$\mathcal{L}^{(k)} e^{sz} = \psi_Z^{(k)}(s) e^{sz}$$

Killed Moment PIDE

Target function:

$$U(t,z) := \mathbb{E}_z^{(k)}\left[g_k(Z_t;h_0)\mathbf{1}_{\{\tau(h_0)>t\}}\right], \qquad z > -h_0$$

Backward Kolmogorov PIDE for the killed process:

$$\partial_t U(t,z) = \mathcal{L}^{(k)} U(t,z), \qquad z > -h_0$$

Initial and absorbing boundary:

$$U(0,z) = g_k(z;h_0), \qquad U(t,z) = 0 \quad \text{for } z \leq -h_0$$

One time variable; one state variable z for the ratio process Z

Laplace Transform Trick

Time-domain PIDE:

$$\partial_t U(t,z) = \mathcal{L}^{(k)} U(t,z), \qquad U(0,z) = g_k(z;h_0)$$

Laplace transform in time:

$$\widehat{U}(q,z) = \int_0^\infty e^{-qt} U(t,z)\, dt, \qquad q > 0$$

Laplace-domain resolvent:

$$(q - \mathcal{L}^{(k)})\widehat{U}(q,z) = g_k(z;h_0), \qquad z > -h_0$$

Sepp [5] idea:

Laplace transform + double-exponential jumps $\rightarrow$ finite algebraic system

Here:

$$\psi_Z^{(k)}(\gamma) = q \quad \Longrightarrow \quad \text{degree-six characteristic equation}$$

Three admissible roots + three boundary conditions $\rightarrow 3 \times 3$ system

Resolvent Solution Form

Resolvent decomposition:

$$\widehat{U}(q, z; h_0) = \widehat{U}_{\mathrm{part}}(q, z; h_0) + \widehat{U}_{\mathrm{hom}}(q, z; h_0)$$

Particular solution:

$$\widehat{U}_{\mathrm{part}}(q, z; h_0) = \sum_{j=0}^{k} \frac{c_{k,j}(h_0)}{q - \psi_Z^{(k)}(j)} e^{jz}$$

Homogeneous solution:

$$\widehat{U}_{\mathrm{hom}}(q, z; h_0) = \sum_{m=1}^{3} B_m^{(k)}(q; h_0) e^{\gamma_{m+3}(q)(z+h_0)}$$

Boundary coefficients:

$$\underbrace{\mathbf{M}_{\mathrm{bar}}^{(k)}(q)}_{3\times 3} \underbrace{\mathbf{B}^{(k)}}_{3\times 1} = -\underbrace{\mathbf{b}^{(k)}}_{3\times 1}$$

PIDE solve $\rightarrow$ summation of exponentials + one 3×3 linear system

Numerical Example

Conditional statistics; Laplace–resolvent vs MC survival

h_0	H_0	$p_{\text{surv}}^{\text{Lap}}$	$p_{\text{surv}}^{\text{MC}}$	Mean	Variance	Skewness	Ex. Kurt.
0.07	1.07	0.3093	0.3206	1.1133	0.0325	0.6550	0.4863
0.10	1.11	0.4879	0.4985	1.0413	0.0306	0.6320	0.3989
0.25	1.28	0.9505	0.9524	0.9278	0.0287	0.2692	−0.1028
0.50	1.65	1.0000	0.9999	0.9573	0.0169	0.0955	0.1592
0.75	2.12	1.0000	1.0000	0.9812	0.0114	0.2026	0.1463
1.00	2.72	1.0000	1.0000	0.9949	0.0092	0.2515	0.1444
1.50	4.48	1.0000	1.0000	1.0095	0.0078	0.2724	0.1428
2.00	7.39	1.0000	1.0000	1.0166	0.0074	0.2705	0.1380
∞	∞	1.0000	—	1.0255	0.0072	0.2627	0.1286

- Smaller H_0: lower survival probability; higher liquidation probability
- Conditional mean: U-shape across H_0; high near boundary, trough around $H_0 \approx 1.3$–1.7
- Conditional variance: decreasing as H_0 increases

Bottom row: no-shorting analytical limit

Numerical Example

Laplace minus MC; wall-clock computation time

h_0	H_0	Δp_{surv}	ΔMean	ΔVar.	ΔSkew.	ΔKurt.	Lap.	MC
0.07	1.07	−0.0113	0.0044	−0.0001	−0.0144	−0.0600	0.15	62.53
0.10	1.11	−0.0105	0.0036	−0.0003	−0.0034	−0.0322	0.15	63.68
0.25	1.28	−0.0019	0.0005	−0.0002	0.0035	−0.0177	0.14	61.30
0.50	1.65	0.0000	−0.0002	−0.0001	−0.0116	−0.0153	0.14	61.51
0.75	2.12	0.0000	−0.0001	−0.0001	−0.0122	−0.0459	0.14	61.07
1.00	2.72	0.0000	−0.0001	−0.0001	−0.0142	−0.0587	0.14	60.75
1.50	4.48	0.0000	−0.0001	−0.0000	−0.0152	−0.0665	0.14	64.36
2.00	7.39	0.0000	−0.0001	−0.0000	−0.0149	−0.0651	0.14	61.99

- $\Delta p_{\mathrm{surv}} < 0$: MC misses intra-period barrier crossings
- Gap largest near $H_0^{\min}$; negligible for larger health buffers
- Runtime: Laplace $\approx$ 0.14s vs MC $\approx$ 62s; about 440$\times$ speed-up

Sizing Objective

Choice variable: $h_0 = \log H_0$; equivalently initial borrow amount

Representative objective:

$$\Phi(h_0) = \mathbb{E}[\Pi_T \mid \tau > T] - \frac{\alpha}{2} \operatorname{Var}(\Pi_T \mid \tau > T) - \delta\, p_{\mathrm{liq}}(h_0),$$

Liquidation probability:

$$p_{\mathrm{liq}}(h_0) = \mathbb{P}(\tau \leq T)$$

One-dimensional search:

$$h_0^\star \in \arg\max_{h_0 > 0} \Phi(h_0)$$

Tradeoff: expected payoff vs variance vs liquidation probability

Interest-Rate Carry: Supply/Borrow Accrual

Collateral earns supply rate r_s; debt accrues borrow rate r_b (Aave indices)

$$V_{1,t} = n_1 S_{1,0} e^{X_{1,t}} e^{r_s t}, \qquad V_{2,t} = n_2 S_{2,0} e^{X_{2,t}} e^{r_b t}$$

Health factor picks up the *net* carry $c = r_s - r_b$ in its drift:

$$H_t = H_0 \exp(Z_t + ct), \qquad h_t = h_0 + Z_t + ct$$

Key point: deterministic carry = a per-leg drift shift

$$\mu_1 \mapsto \mu_1 + r_s, \qquad \mu_2 \mapsto \mu_2 + r_b$$

- Every killed moment $\mathbb{E}[\Pi_T^k]$ and $p_{\mathrm{surv}}(h_0)$ apply verbatim at the shifted drifts
- No new analysis: same 3×3 resolvent, same Talbot inversion
- Positive net carry $\Rightarrow$ first-order buffer that *raises* p_{surv} at fixed h_0

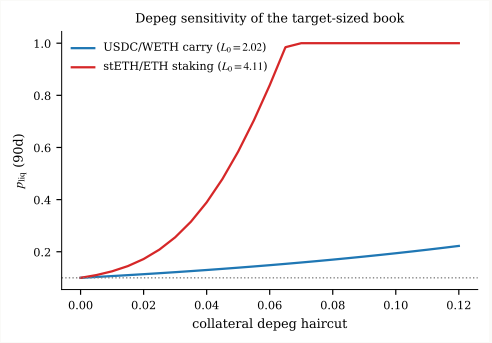
Calibrated to live Aave v3 Ethereum rates (`get_aave_hourly.py`, 2026-08-04)

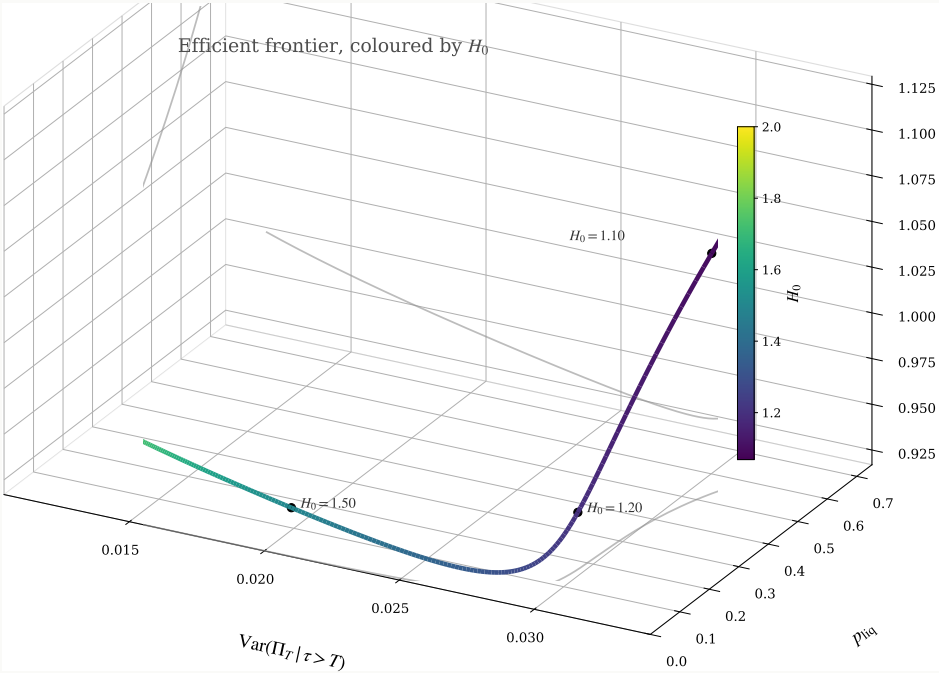
Carry Books and Depeg Fragility

Two Aave v3 books sized to $p_{\mathrm{liq}}^{\dagger} = 10\%$; collateral = asset 1, debt = asset 2

Book	carry	$H_0^{\dagger}$	$L_0^{\dagger}$	$\mathbb{E}[\Pi_T \mid \tau > T]$	cond. Sharpe
USDC/WETH ($c = +1.18\%$)	none	1.547	2.017	1.0710	4.62
	+carry	1.543	2.022	1.0840	4.62
stETH/ETH ($c = +0.49\%$)	none	1.070	4.110	1.0849	2.18
	+carry	1.070	4.114	1.0890	2.18

- Correlation, not carry, sets admissible leverage: collinear stETH/ETH tolerates $L_0 \approx 4.1$; independent USDC/WETH capped near 2.0
- Carry shifts the frontier favourably but modestly at these horizons
- Depeg fragility: a d haircut = same book at $h_0 + \log(1-d)$
- stETH/ETH (4.1×): 5% haircut → $p_{\mathrm{liq}} = 58\%$; 10% → certainty
- USDC/WETH (2.0×) degrades gracefully: 10% → 19%




Efficient frontier, coloured by H_0
$H_0 = 1.10$
$H_0 = 1.20$
$H_0 = 1.50$
H_0
2.0
1.8
1.6
1.4
1.2
1.125
1.100
1.075
1.050
1.025
1.000
0.975
0.950
0.925
0.015
0.020
0.025
0.030
$Var(\Pi_T \mid \tau > T)$
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
p_{liq}

Conclusions and Discussion

- DeFi long–short $\rightarrow$ collateral/debt first passage
- Central sizing variable: $h_0 = \log H_0$
- Kou jump diffusion $\rightarrow$ finite-dimensional semi-analytic moments
- Fast liquidation-aware sizing; no repeated Monte Carlo
- Interest-rate carry $\rightarrow$ per-leg drift shift; interest-bearing books sized with no re-derivation

Ongoing developments

- Liquidation recovery and dynamic collateral management
- *Stochastic* borrowing- and lending-rate modelling
- Oracle-delay and execution-slippage overlays

Thank you

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